

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat
Department of Computer Science and Engineering
B.Tech. II - Semester-III
CS231 – Object Oriented Programming Practical Assignment
Lab Assignment-3 Assignment: Object, Class & Function Overloading

1. Write a C++ program to access public members of a class, by using the (.)dot operator.
These are members marked with a public access modifier.
Example: Price of Car A1 = 10000
Price of Bike A2 = 2000
Number of Car available in showroom = 4
Number of Bike available in showroom = 3
2. Write a C++ program to create an object of a class and access class attributes
Example: Student's Roll No.: 101
Student's Name: First_Name Last_Name
Student's Percentage: 90.4
3. Write a C++ program to calculate the area of a circle using an area function where radius is a private and area function is a public member of a class circle.
4. Develop a Library class that contains a collection of books (Book objects). Implement methods to add a book, remove a book, and display all books.
5. Create a BankAccount class with attributes accountNumber, accountHolder, and balance. Implement methods to deposit and withdraw money. Ensure to display an appropriate message if a withdrawal amount exceeds the available balance.
6. Create a C++ program that demonstrates function overloading using a class. You are required to design a class named Calculator that performs addition operations. The add() function should be overloaded to handle the following cases:
 - Add two integers
 - Add three integers
 - Add two floating-point numbers
 - Add a double and an integer
7. Design a class GradeCalculator with overloaded functions calculateGrade() that return the grade of a student based on:
 - Percentage marks (float)
 - Marks out of 100 (int)
 - Marks in 5 subjects (int, int, int, int, int)Use simple grading logic:
A: 90–100, B: 80–89, C: 70–79, D: 60–69, F: <60
8. Create a class Distance Converter that uses function overloading to convert distances:
 - From kilometers to meters: convert(float kilometers)
 - From meters to centimeters: convert(int meters)
 - From miles to kilometers: convert(double miles, char unit)Each overloaded function should return the converted distance.
9. Develop a simple smart home system that allows a user to manage one or more smart devices (e.g., Light).

- a. **Task:**
 - i. **Design a Smart Device Class:** Implement a C++ class to represent a smart device.
 - ii. **Define Member Variables and Functions:**
 - 1. The class should have member variables to **store the device's state** (e.g., status, brightness).
 - 2. Implement member functions to **manipulate** and **access** the device's state.
- b. **System Requirements:** Your developed system must fulfil the following requirements,
 - i. **Implement Function Overloading:** Use function overloading for one or more actions to provide different ways of controlling the device.
 - ii. **Display Device Status:**
 - 1. Your program must display the initial state of the device.
 - 2. After applying various settings and actions, the program should display the updated state.
- c. **For example:** Implement a SmartLight device with the following properties,
 - i. *Name:* A string to identify the light (e.g., "Living Room Light").
 - ii. *Status:* A boolean or enum to represent its on/off state.
 - iii. *Brightness:* An integer value ranging from 0 to 100, where 0 means the light is off.
 - iv. **Functions:** *togglePower()*, *changeSetting(int brightness)*, *displayState()*
- d. **Extend Functionality:** After successfully implementing the SmartLight, implement the same requirements for a **second smart device of your choice**.